

SmartGym Adaptive Training System

SquareLine Handoff

UI handoff notes for SquareLine/LVGL assets.

Version 1.0 Prototype

ESP32-S3 firmware | Firebase RTDB | React dashboard

Generated documentation package

SmartGym Adaptive Training System

SquareLine Handoff

UI handoff notes for SquareLine/LVGL assets.

Version 1.0 Prototype

ESP32-S3 firmware | Firebase RTDB | React dashboard

Generated documentation package

SmartGym UI Handoff (SquareLine)

Hey! You have full design freedom.

Don?t worry about matching current LVGL layout, I?ll adapt code to your IDs.

What this app does

Gym machine UI with:

- live coaching while training
- rest countdown between sets
- session summary at the end

Screens we need

1. Idle
2. Main Training
3. Rest Overlay (modal on top of Main)
4. Summary
5. Calibration Gate
6. Profile
7. Debug/Service (can be basic)

Important: only one start/finish control in Main ('START' / 'FINISH' toggle).

Data you can use on UI (descriptive)

Live

- ****Selected weight (kg)****: current load chosen for the set.
- ****Suggested load seed (kg)****: baseline recommended weight before live adjustments.
- ****Weight factor****: multiplier used to scale suggested load for user/goal context.
- ****Current set / target sets****: progress through the workout plan (example '2 / 4').
- ****Reps in set / target reps****: progress inside the active set (example '7 / 10').
- ****ROM %****: current range of motion depth as a percent (0-100).
- ****User ROM bounds****: personalized range limits ('userBottomPct' to 'userTopPct').
- ****Velocity (%/s)****: movement speed normalized as percent per second.
- ****Rep phase****: live phase label ('LIFT', 'LOWER', 'HOLD', 'REST').
- ****Quality state****: fast coaching state ('READY', 'SLOW', 'SPEED OK', 'TOO FAST', etc.).
- ****Session timer****: elapsed workout time in the active session.
- ****Rest countdown****: seconds remaining before the next set.

- **Rest target ('restSeconds')**: planned rest time for the current exercise/goal.
- **Rep timing (ms)**:
- 'riseMs': concentric/lift duration
- 'lowerMs': eccentric/lower duration
- 'topPauseMs': pause near top position
- 'bottomPauseMs': pause near bottom position

Motion guide (important)

- **Ideal motion curve**: target path for a good rep over time.
- **Live dot position**: athlete's current movement point plotted on that graph.
- **Live halo**: visual emphasis around the live dot so it's easy to follow quickly.
- **On-path state**: whether live movement is close enough to the ideal path.
- **ROM target band**: highlighted target ROM zone (single fixed chart mode, no expanded graph).

Session end / summary

- **Quality score + tier**: overall session quality (0-100) and readable tier label.
- **Sets completed**: completed sets vs planned sets.
- **Valid / invalid reps + valid rate**: accepted reps, rejected reps, and success percentage.
- **Average ROM %**: average ROM on valid reps.
- **Average velocity (%/s)**: average rep speed on valid reps.
- **Volume load (kg)**: total effective load ('selected weight * valid reps').
- **Duration**: total session time ('mm:ss' or similar).
- **Sync/storage status**: whether saved locally, queued, or synced.

Extra useful values you can show (already available)

- **Best ROM %**: best rep ROM reached in the session.
- **Average eccentric velocity (%/s)**: lowering-phase speed quality signal.
- **Fast eccentric warnings**: count of reps lowered too quickly.
- **Total rest time**: total rest accumulated in session.
- **Average rest per set**: pacing indicator.
- **Rep quality trend**: first sets vs last sets quality change.
- **Fatigue ROM drop**: ROM loss from early to late sets.
- **Fatigue velocity drop**: speed loss from early to late sets.
- **ROM compliance rate**: percent of valid reps that met user ROM target.
- **Ideal ROM hit rate**: percent of valid reps that met machine ideal ROM target.
- **Invalid reason breakdown**:

- short ROM
- too fast
- top not reached
- no concentric phase

Data keys to include in UI mapping (important)

- 'restSeconds'
- 'riseMs'
- 'lowerMs'
- 'topPauseMs'
- 'bottomPauseMs'
- 'suggestedLoadSeedKg'
- 'weightFactor'
- 'userBottomPct'
- 'userTopPct'

Extra coaching metrics available

- ****ROM compliance rate****: percent of valid reps that reached user ROM target.
- ****Ideal ROM hit rate****: percent of valid reps that reached machine ideal ROM target.
- ****Fatigue ROM drop****: how much ROM decreased from early sets to late sets.
- ****Fatigue velocity drop****: how much speed decreased from early sets to late sets.
- ****Invalid reason counts****: rejection reason totals ('short ROM', 'too fast', 'top not reached', etc.).

Update rhythm (so layout expectations are clear)

- Live values (ROM, velocity, phase, quality, timers, rep counters): update continuously during training.
- Rest values: update each second during rest countdown.
- Summary values: update once at session finish, then remain stable on summary screen.

Profile screen (show + edit)

Show

- Card UID
- Name
- Age
- Weight (kg)
- Height (cm)

- Goal
- Gender

Editable

- Name (text input + keyboard)
- Age ('- / +')
- Weight ('- / +', 0.5 kg step)
- Height ('- / +', 1 cm step)
- Goal (prev/next)
- Gender (prev/next)

Save/Cancel behavior

- 'Save': keep changes and return
- 'Cancel': discard changes and return

Validation ranges

- Age: 12 to 90
- Weight: 35.0 to 220.0 kg
- Height: 120 to 230 cm

Profile UX note

- Goal and Gender can be dropdown menus (preferred).

Debug/Service screen requirements

Keep it practical and fast to use. Styling can be simple.

Must show

- Debug status text block
- Hardware status text block

Must include controls

- Machine selector (dropdown)
- ROM injector slider
- Auto rep button
- short press: one cycle
- long press/hold: continuous
- release: stop continuous mode
- Motion reset button
- Jump ROM to top button
- Jump ROM to bottom button

- Cloud sync button
- Sensor mode toggle button
- Log level button
- Encoder buttons:
 - set 0cm
 - set 100cm
 - apply
 - reset
- Quick user buttons:
 - user 1
 - user 2
 - new user
 - anonymous
- Close button

Debug screen intent

- quick hardware checks
- simulation for demos/testing
- calibration and cloud verification
- user/session test shortcuts

What to send me back

1. '.slp' project
2. exported LVGL files
3. fonts/images/icons used
4. small mapping table:

screen	widget_id	meaning
---	---	---
Main	btn_start_finish	start_or_finish
Main	lbl_velocity	live velocity

```
| Summary | lbl_summary_main | summary metrics block |
```

That's it. I'll handle the integration.

Quick event list (so wiring is easy)

Please include widgets for these actions:

- 'start_or_finish'
- 'weight_minus_large' ('-5')
- 'weight_minus' ('-2.5')
- 'weight_plus' ('+2.5')
- 'weight_plus_large' ('+5')
- 'start_calibration'
- 'rest_skip'
- 'open_profile'
- 'open_service'
- 'profile_save'
- 'profile_cancel'
- 'calibration_gate_calibrate'
- 'calibration_gate_skip'

When each screen appears

- 'Idle': no active workout / waiting for user
- 'Main Training': active training flow
- 'Rest Overlay': between sets during rest countdown
- 'Summary': right after finishing session
- 'Calibration Gate': when user needs calibration decision
- 'Profile': when user edits profile
- 'Debug/Service': service/testing tools

Fill-this template (copy/paste)

```
| screen | widget_id | type | meaning |
```

```
| --- | --- | --- | --- |
```

```
| Main | | button | start_or_finish |
```



```
| Main | | button | weight_minus_large |
```

```
| Main | | button | weight_minus |
```

```
| Main | | button | weight_plus |
```

```
| Main | | button | weight_plus_large |
```

```
| Main | | label | live_rom |
```

```
| Main | | label | live_velocity |
```

```
| Main | | chart | ideal_curve + live_dot |
```

```
| Rest Overlay | | button | rest_skip |
```

```
| Summary | | label | summary_main_metrics |
```

```
| Summary | | label | summary_detail |
```

```
| Calibration Gate | | button | calibration_gate_calibrate |
```

```
| Calibration Gate | | button | calibration_gate_skip |
```

```
| Profile | | dropdown | goal |
```

```
| Profile | | dropdown | gender |
```

```
| Profile | | button | profile_save |
```

```
| Profile | | button | profile_cancel |
```